

JPR Assignment 4

1. WAP to create three buttons with caption OK, RESET and CANCEL.

Ans.

```
import java.awt.*;
class MyFrame extends Frame
{
    Button ok, reset, cancel;
    MyFrame(string s)
    {
        super(s);
        setLayout(null);
        setVisible(true);
        setSize(500,300);
        ok = new Button("OK");
        reset = new Button("RESET");
        cancel = new Button("CANCEL");
        ok.setBounds(50,50,50,50);
        reset.setBounds(100,50,100,50);
        cancel.setBounds(100,50,100,50);
        add(ok);
        add(reset);
        add(cancel);
    }
    public static void main(String [] args)
    {
        MyFrame f = new MyFrame("Buttons");
    }
}
```

2. Define Layout Manager. Explain Grid Layout with example.

Ans. i. Layout means the arrangement of components within the container.

ii. The task of laying out the controls are done automatically by the Layout Manager.

iii. The Layout manager is set by the setLayout() method.

iv. Layout Manager are founded in java.awt package for displaying the graphical components.

Grid Layout :-

i. Grid Layout lays out components in a two-dimensional grid.

ii. The number of rows and columns is defined when the grid layout manager is created.

iii. All components have same size in grid layout.

Example :-

```
import java.awt.*;
public class GridLayoutDemo extends Frame
{
    GridLayoutDemo()
    {
        setTitle(GridLayoutDemo);
        setSize(500,500);
        setVisible(true);
        setLayout(new GridLayout(3,2,3,3));
        Button b1 = new Button("1");
        add(b1);
        Button b2 = new Button("2");
        add(b2);
        Button b3 = new Button("3");
        add(b3);
        Button b4 = new Button("4");
        add(b4);
        Button b5 = new Button("5");
        add(b5);
        Button b6 = new Button("6");
        add(b6);
    }
    public static void main(String [] args)
    {
        GridLayoutDemo d = new GridLayoutDemo();
    }
}
```

3. WAP to display JComboBox that will have following items :- Pen, Pencil, Sharpener, Eraser, Notebook.**Ans.**

```
import javax.swing.*;
public class JFrameDemo extends JFrame
{
    public JFrameDemo()
    {
        String items[] = {"Pen", "Pencil", "Sharpener", "Eraser", "Notebook"};
        JComboBox cb = new JComboBox(items);
        cb.setBounds(50,50,90,20);
        add(cb);
        setLayout(null);
    }
}
```

```
setSize(400,500);
setVisible(true);
}
public static void main(String [] args)
{
    new JFrameDemo();
}
}
```

4. List AWT Controls.

Ans. AWT controls are GUI components used to accept input or display output in a Java application. List of AWT Controls are as follows :-

1. Label
2. Button
3. TextField
4. TextArea
5. CheckBox
6. CheckBoxGroup

5. Enlist Swing Features.

Ans. The features of Swing are as follows :-

1. Light weight
2. Rich controls :- Rich set of advanced controls like Tree, slider, colour, etc.
3. Borders :- Can draw borders in different style using `setBorder()`.
4. Easy mouse less operation :- Easy to connect
5. Easy scrolling
6. Pluggable look and feel.